Experiment 14

Introduction to Deluge Scripting: Writing Custom Functions and Basic Programs

1. Aim

To understand basic Deluge syntax and use simple scripts or custom functions to extend a Zoho Creator application.

2. Objectives

- Recognize basic Deluge statements, variables, conditions, and data types.
- Write a simple conditional rule for validating or calculating a value.
- Understand how scripts can respond to form events or support custom functions.
- Test scripts with normal, boundary, and invalid inputs.

3. Requirements

Access to Zoho Creator or Deluge documentation, a test application, and sample records.

4. Theory

Deluge (Data Enriched Language for the Universal Grid Environment) is a scripting language used across Zoho products to add logic to applications and workflows. A script may assign values, evaluate conditions, perform calculations, or interact with records according to the context and permissions. For example, a request application might calculate a total cost from quantity and unit price or flag a request that exceeds a defined threshold. Syntax and available tasks depend on the Zoho product and the place where the script runs. Scripts should validate input, handle missing values, and avoid exposing sensitive data.

5. Procedure

1. Open the scripting or workflow editor available in the test application.
2. Review the editor's supported context and a basic Deluge example from official documentation.
3. Create a small script, such as calculating quantity multiplied by unit price.
4. Add a conditional statement to flag values above a chosen threshold.
5. Run the script with normal, zero, boundary, and missing-value test cases.
6. Check the resulting field values or messages and correct any errors.
7. Document what the script does and where it is triggered.

6. Case Study and Analysis

A laboratory request form records quantity and unit price for requested components. Staff previously calculated the estimated total manually, sometimes entering inconsistent values. A simple Deluge script calculates the estimated total and flags requests above a departmental threshold for additional review. The script should first check that quantity and unit price are present and valid, then calculate the total. A conditional rule can set a review indicator when the total exceeds the configured limit. During testing, the developer checks a normal request, a zero quantity, a high-cost request, and missing input. The team verifies that the threshold is based on current policy and that users cannot change a calculated value in a way that bypasses approval. The script is kept small and clearly documented so another maintainer can understand it. More complex applications should separate reusable logic into functions where appropriate and follow the platform's current Deluge syntax and execution-context rules. The example demonstrates how a little scripting can improve consistency while leaving the form and workflow visually configurable.

7. Result / Discussion

A basic Deluge logic example can automate a calculation and flag records that meet a condition. Testing helps identify missing-input and boundary-case issues.

8. Conclusion

Deluge extends low-code applications with custom logic. Reliable scripts use clear conditions, validate inputs, and are tested against realistic edge cases.

9. Precautions / Best Practices

- Use fictional or non-sensitive test data during practice.
- Verify settings and results before using the application or configuration in a real environment.
- Record observations clearly and follow the platform's current documentation, security, and access-control guidance.